Spatial weights matrix
Spatial statistics

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What for do we need the spatial weights matrix?

- **It is to incorporate spatial dependence into the model - always a question „how”:**
  - What are a good criteria to define closeness in space
  - Which other units should be treated as neighbours (who influence given location)
  - Should W be selected *a priori ?* – then it is about the assumption on the nature of links
  - Should W be derived from the data and tested ? – then it plays the role of the filter

Functions of W

- **Spatial weights matrix W**
  - Different criteria of neighbourhood and the strength of the links
  - It indicates the relative location of units (with regard to each other) – the structure of neighbourhood
  - assigned weights are to measure the intensity of the relationship among spatial units
  - Allows for computing „spatial lag” – average of values in neighbourhood
  - It is used in calculations of spatial statistics and estimation of models
  - It is crucial in measuring the externalities
  - It may be symmetric, but not necessarily

W construction

Spatial weights matrix W

- size: *n* × *n*,
- diagonal w_{ii} =0
- it may be symmetric, but not necessarily
- ***wij*** - weights for each pair of locations; result from a rule on neighbourhood structure, define the existence and strength of the spatial relations among locations

$$\mathbf{W} = \begin{pmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & \cdots & w_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{n1} & w_{n2} & \cdots & w_{nn} \end{pmatrix}$$

For most frequently used – **contiguity matrix**:

$$w_{ij} = \begin{cases} 1 & \text{if } i \text{ and } j \text{ are contiguous} \\ 0 & \text{if } i \text{ and } j \text{ are not contiguous} \end{cases}$$

For also often used **inverse distance**:

$$w_{ij} = \begin{cases} \frac{1}{d_{ij}^{\alpha}} & \text{if } i \neq j \\ 0 & \text{if } i = j \end{cases}$$

Also possible: **negative exponential**

$$w_{ij} = \exp\left(-\frac{d_{ij}}{\alpha}\right)$$

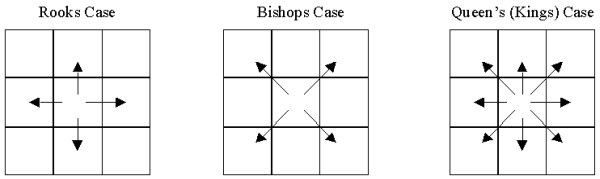
Matrix of neighbourhood and matrix of spatial weights

- **Spatial neighbourhood matrix**

$$\tilde{W} = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

- **Spatial weights matrix**

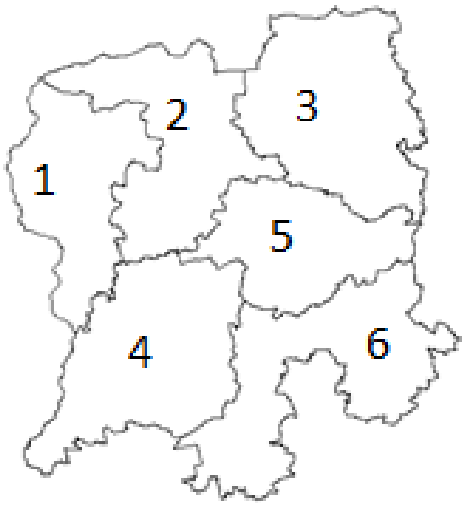
$$W = \begin{pmatrix} 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \end{pmatrix}$$



- Spatial relation of areas, different criteria possible*, binary structure of neighbourhood
  - contiguity (share the border)
  - K nearest neighbours
  - within the radius of d km (great circle)
  - all with all other regions, but of inverse distance
- Strength of spatial relations - binary structure changed to weights:
  - $w_{ij}=1/n$ for n neighbours (row-standardised matrix to one
  - $w_{ij}=1/dist_{ij}$ for inverse distance

* Literature also offers for regular shapes other criteria: *rook* (common border), *bishop* (common vertex), *queen* (either common border or vertex)

Example of W (1)



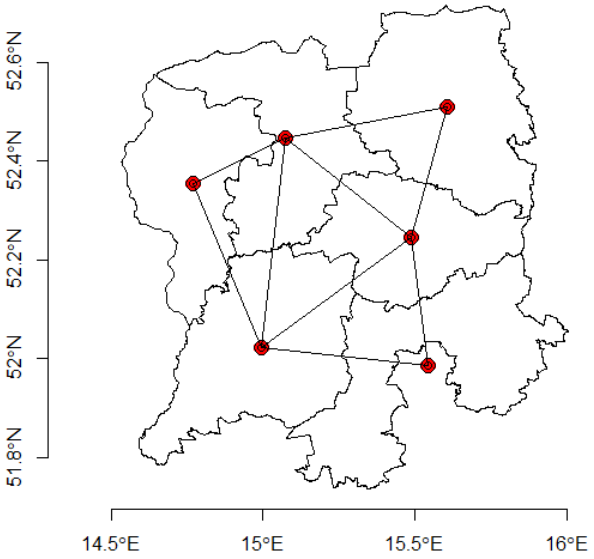
Neighbourhood structure

$$W = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

Matrix of spatial weights

$$W = \begin{pmatrix} 0 & 1/2 & 0 & 1/2 & 0 & 0 \\ 1/4 & 0 & 1/4 & 1/4 & 1/4 & 0 \\ 0 & 1/2 & 0 & 0 & 1/2 & 0 \\ 1/4 & 1/4 & 0 & 0 & 1/4 & 1/4 \\ 0 & 1/4 & 1/4 & 1/4 & 0 & 1/4 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix}$$

Example of W (2)



Distances calculated bewteen centroids (usually Euclidean distance)

Distances between regions

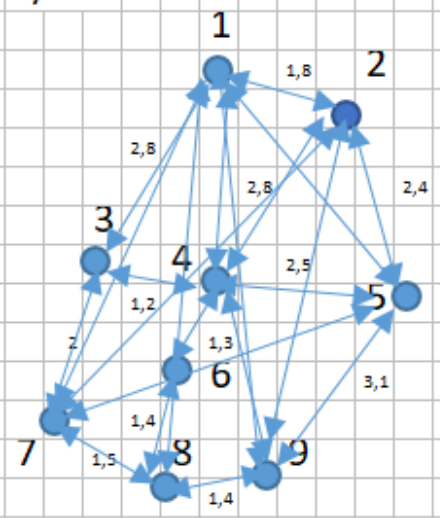
	Reg1	Reg2	Reg3	Reg4	Reg5	Reg6
Reg1	0,00					
Reg2	0,78	0,00				
Reg3	0,40	0,85	0,00			
Reg4	0,43	0,54	0,32	0,00		
Reg5	0,54	0,29	0,73	0,46	0,00	
Reg6	0,55	0,53	0,86	0,66	0,27	0,00

Inverse distance matrix

	Reg1	Reg2	Reg3	Reg4	Reg5	Reg6
Reg1	0,00	1,28	2,50	2,33	1,85	1,82
Reg2	1,28	0,00	1,18	1,85	3,45	1,89
Reg3	2,50	1,18	0,00	3,13	1,37	1,16
Reg4	2,33	1,85	3,13	0,00	2,17	1,52
Reg5	1,85	3,45	1,37	2,17	0,00	3,70
Reg6	1,82	1,89	1,16	1,52	3,70	0,00

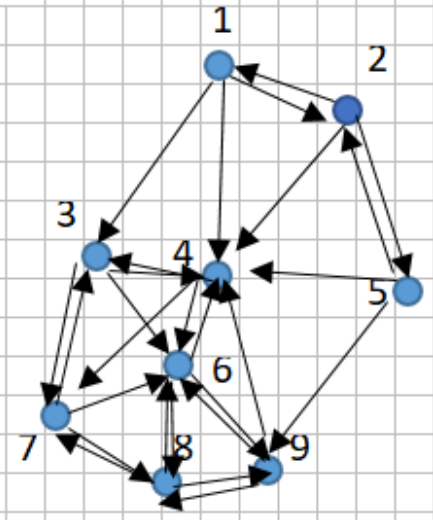
Summary of W

1/dist

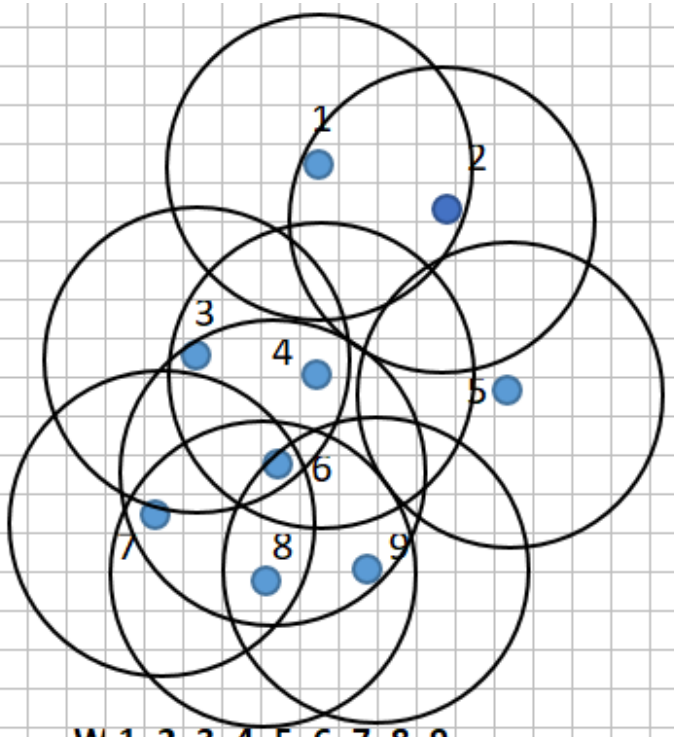


W	1	2	3	4	5	6	7	8	9
1	0	…	…	…	…	…	…	…	…
2	…	0	…	…	…	…	…	…	…
3	…	…	0	…	…	…	…	…	…
4	…	…	…	0	…	…	…	…	…
5	…	…	…	…	0	…	…	…	…
6	…	…	…	…	…	0	…	…	…
7	…	…	…	…	…	…	0	…	…
8	…	…	…	…	…	…	…	0	…
9	…	…	…	…	…	…	…	…	0

knn=3



W	1	2	3	4	5	6	7	8	9
1	0	1	1	1	0	0	0	0	0
2	1	0	0	1	1	0	0	0	0
3	0	0	0	1	0	1	1	0	0
4	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	0	0	0	1
6	0	0	0	1	0	0	0	1	1
7	0	0	1	0	0	1	0	1	0
8	0	0	0	0	0	1	1	0	1
9	0	0	0	1	0	1	0	1	0



W	1	2	3	4	5	6	7	8	9
1	0	1	0	0	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0
3	0	0	0	1	0	1	0	0	0
4	0	0	1	0	0	1	0	0	0
5	0	0	0	0	0	0	0	0	0
6	0	0	1	1	0	0	1	1	1
7	0	0	0	0	0	1	0	1	0
8	0	0	0	0	0	1	1	0	1
9	0	0	0	0	0	1	0	1	0

Summary of terms

- **W** – spatial weights matrix, transformed neighbourhood matrix

- **Spatial lag** – average of values in neighbourhood
  →one adds spatial lags of **y, x** and **e** into models

- **Spatial autocorrelation** – relation between observation in location *i* and observation in location *j*

- **Spillover effects –** Direct impact ΔX(**i**) → ΔY(**i**) and indirect impact ΔX(**i**) → ΔY(**j**)

Exploratory Spatial Data Analysis (ESDA)

ESDA - Exploratory Spatial Data Analysis → *to find structure (cluster, unusual observations) in large and not necessarily homogeneous spatial data sets*

Global measures (one value for whole dataset)

- Moran's I (1950) (more sensitive to outliers)
- Geary's C (1954) (more sensitive to local autocorrelation)
- Getis-Ord's G (1992)

Global spatial autocorrelation (spatial dependence): positive autocorrelation means that locations close to each other exhibit more similar values than those further apart. High (low) values are sistematically surrounded by high (low) values. This should be true for the whole territory.

Local measures - LISA – Local Indicator of Spatial Association (separate values for each spatial unit)

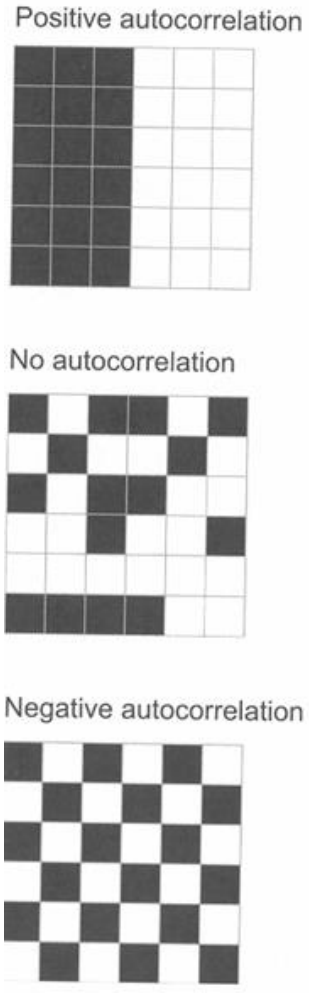
- Local Moran's I (Anselin 1995)
- Local Geary's C (Anselin 1995)
- Local G* (Getis and Ord, 1995)

Local spatial autocorrelation: if spatial relation does not occur systematically (no global spatial dependence) we can look for some local patterns, which are typical for some clusters only

- We can also get Moran Scatterplot and maps of LISA

Spatial autocorrelation (1)

Spatial autocorrelation



- Similar values neighbouring more often than in random distribution, when extreme then club or spatial heterogenity
- When no clear spatial pattern is visible
- When similar values touch similar values more rarely then in random distribution

What does spatial autocorrelation mean? (1)

When tracking for spatial autocorrelation one should be aware of the consequences of this phenomenon:

- **Null Hypothesis: no spatial autocorrelation**
  - Spatial randomness (vs. spatial clustering)
  - Values observed at a location do not depend on values observed at neighbouring locations
  - Observed spatial pattern of values is the same as any other spatial pattern
  - The location of values may be altered (spatial permutation) without affecting the information content of the data
- **Alternative Hypotheses: spatial dependence**
  - Positive spatial autocorrelation: similar values tend to cluster in space (neighbours are similar)
  - Negative spatial autocorrelation: Neighbours are different

What does spatial autocorrelation mean? (2)

In case of lack of spatial randomness, there are few types of spatial association possible:

- Spatial autocorrelation: similar (dissimilar) values in space tend to cluster together
- Spatial heterogeneity: spatial regimes, space is not homogeneous, there may exist clubs of similar values inside, but very different from outside values

Spatial autocorrelation (2)

Moran's I - ***statistic for spatial autocorrelation***

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij}(X_i - \bar{X})(X_j - \bar{X})}{\sum_i (X_i - \bar{X})^2}$$

- developed by Moran (1950)
- a measure of spatial autocorrelation, for how in general high values are located close to high values (and low next to low)
- interpretation similar to Pearson r correlation
- possible values: -1< Moran I <1
  - when positive I>0, positive spatial autocorrelation
  - when I=0, no spatial autocorrelation
- the expected value for spatial randomness is -1/(N-1)
- Theoretical variance can be computed with normal or random approximation
- standardized statistic approaches N(0,1) → $Z = \frac{I - E(I)}{\sqrt{Var(I)}} \sim N(0,1)$

Spatial autocorrelation (3)

Geary's c - ***statistic for spatial autocorrelation***

$$c = \left(\frac{N-1}{2\sum_i \sum_j w_{ij}}\right) \cdot \left(\frac{\sum_i \sum_j w_{ij}\,(x_i - x_j)^2}{\sum_i (x_i - \bar{x})^2}\right)$$

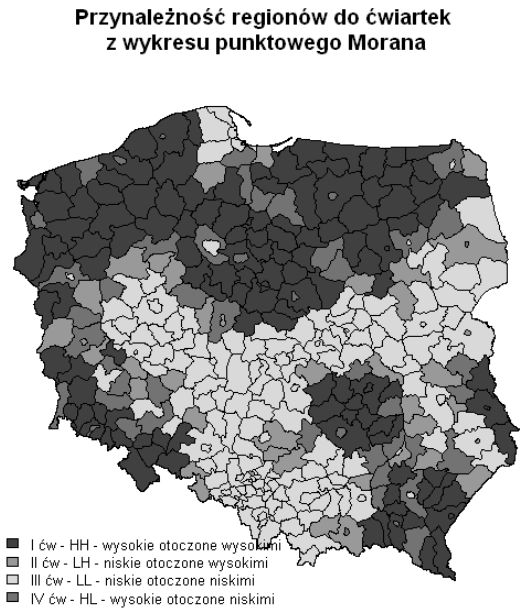
- developed by Geary (1954)
- a measure of spatial autocorrelation, for how in general high values are located close to high values (and low next to low)
- sensitive to local patterns $|x_i - x_j|$
- possible values:
  - for 0<c<1, positive spatial autocorrelation (cooperation)
  - for 1>c>2 negative spatial autocorrelation (competition)
  - for c=1, no spatial autocorrelation (independence)

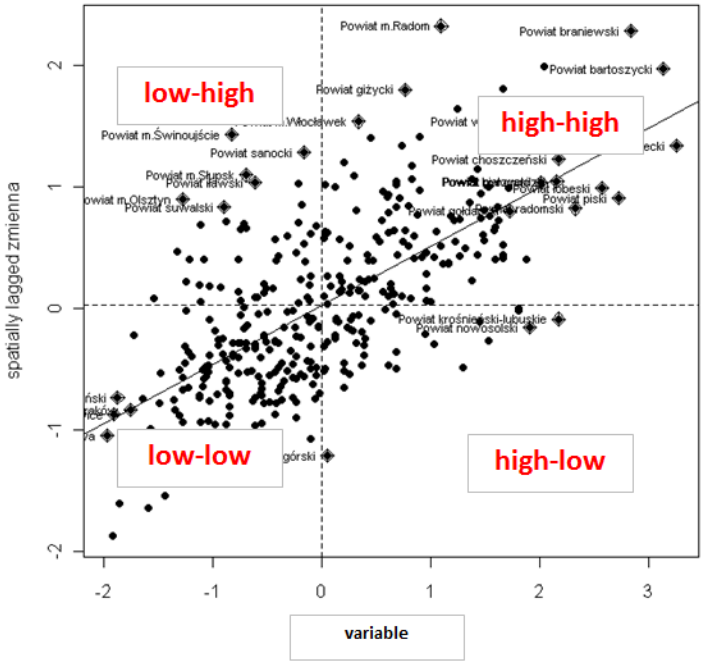
Spatial autocorrelation (3)

- **Moran scatterplot**
  - X: variable of interest
  - Y: spatial lag of X

HH, LL – spatial clusters

HL, LH – spatial outliers





- It is useful in tracking for spatial regimes
- Color of region: „**me vs. around**”
- HH (high-high) (dark colour) – me and my neighbours have high level of phenomenon

Moran scattererplot – example (1)

Let's compare two periods

Method for spatial data:

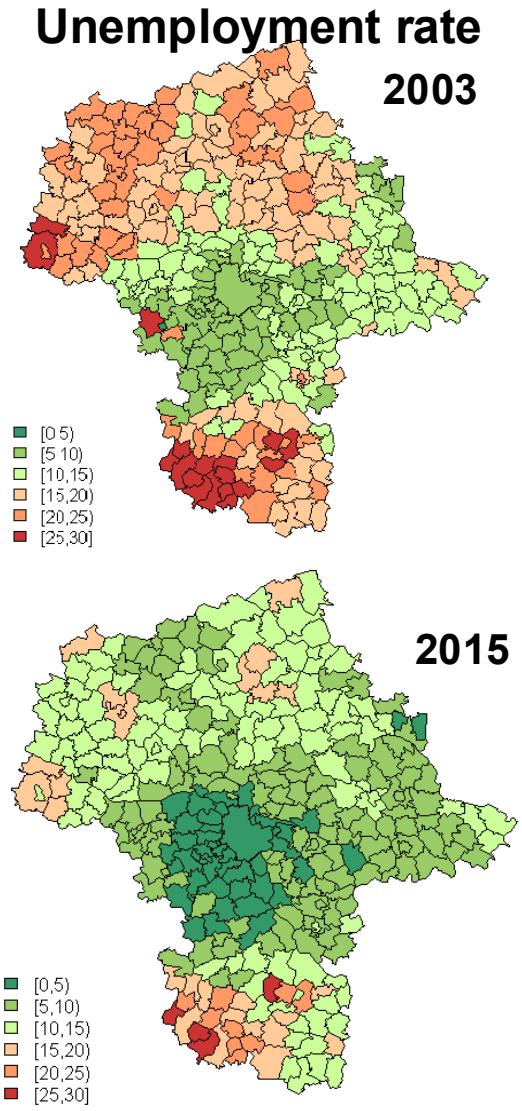
- Observed variable plotted on map for two (or more) time periods
- Proper number of bins and colors from palette selected

Hypotheses:

- Is the spatial distribution of examined variable stable over time?
- What is the trend over time?

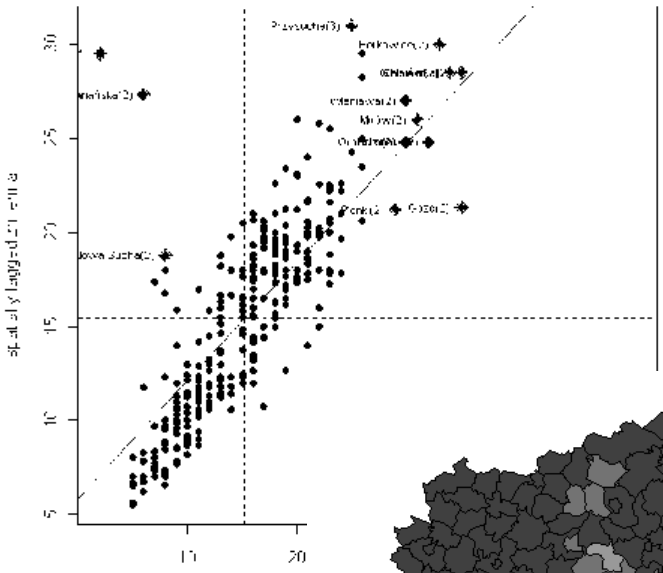
Problem:

- Poor comparisons due to scale changes

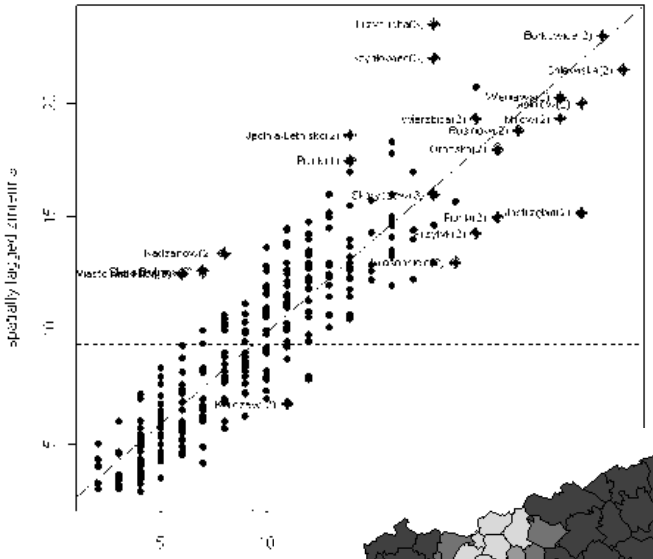


Moran scattererplot – example (2)

Using Moran Scatterplot earses the scale of phenomon – we stay with patial pattern only

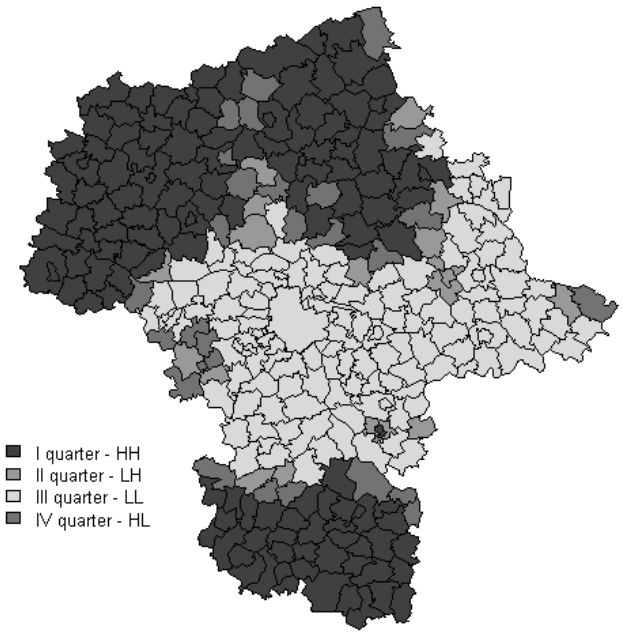


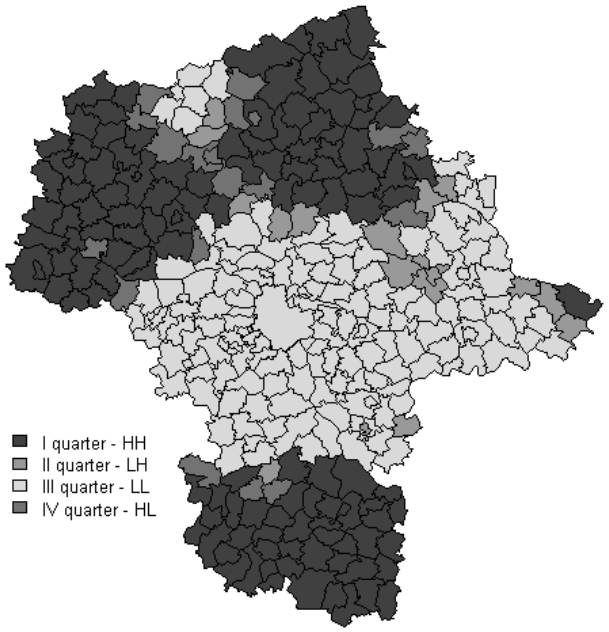
2003



2015

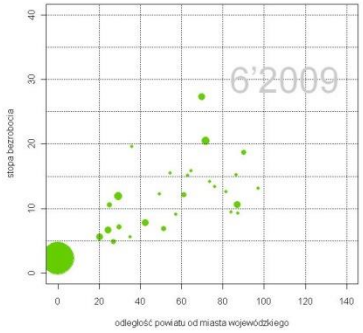
We can see the structural changes in regions.

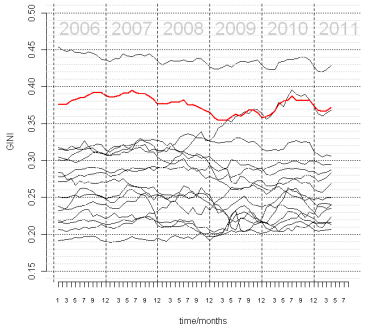


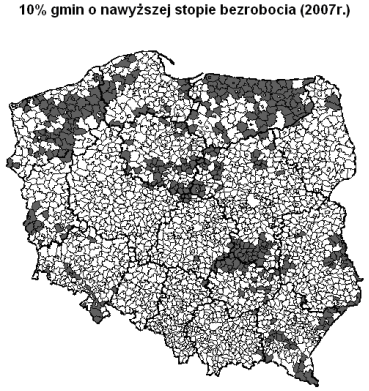


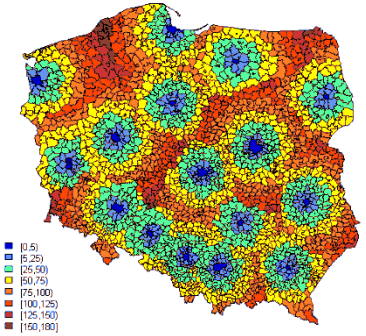
What else we can do with spatial data to get new information? (1)

- **Except Moran statistics** (and other ESDA stats) one can explore:
  - relation of phenomenon (y) to distance (x)
  - cointegration of time series data for each region
  - maps of extreme 10% of values
  - phenomenon in relation to core-periphery pattern



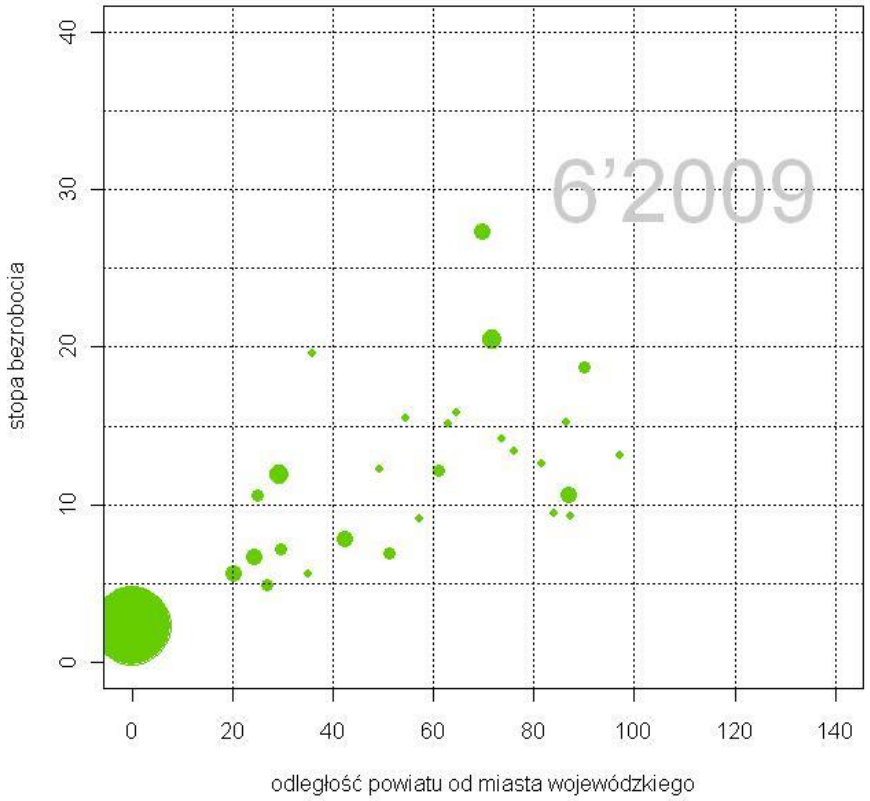




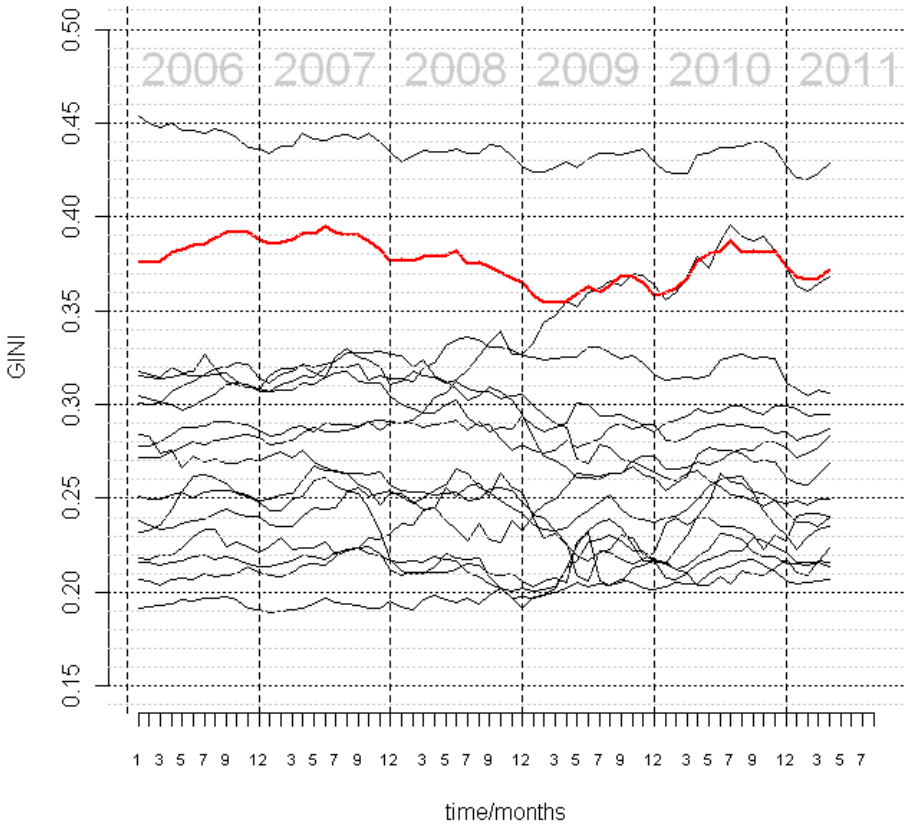


What else we can do with spatial data to get new information? (2)

- Distance-dependent unemployment



- Gini of unemployed persons in NUTS2 regions



What else we can do with spatial data to get new information? (3)

Unemployment rate



Own revenues per capita



Health care system spendigs per capita



What else we can do with spatial data to get new information? (4)

Business units per capita

?

Distance between NUTS5 units and core city

